

ICData RESTful Web Services Technical User Guide

National Institutes of Health/Office of Extramural Research



Electronic Research Administration
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1.1. 1. General Information

1.1.1. 1.1. Purpose

The purpose of this document is to provide a general description and usage guide for the ICData RESTful web services.

1.1.2. 1.2. General Description

IC Extension PL/SQL packages were migrated to RESTful web services. Following table details the PL/SQL packages and corresponding RESTful web services.

No	Package Name	Procedure Name	Web Service
1	ice_dbms_fedp_a_pkg	assign_proc	/ice/fed-person-involvement/assign/{applId}
2	ice_dbms_icpcc_a_pkg	assign_ic_pcc_proc	/ice/ic-pcc/assign/{applId}
3	ice_dbms_icpcc_a_pkg	assign_main_pcc_proc	/ice/main-pcc/assign/{applId}
4	ice_dbms_awfd_a_pkg	assign_proc	/ice/awd-fundings/assign/{applId}

5	ice_dbms_appd_a_pkg	assign_proc	/ice/appd-fundings/assign/{applId}
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Note: The user should not call the list of the above PL/SQL packages but instead call the Web Service methods.

1.1.3. 1.3. Security

A valid certificate is required to access the web service and the mapped user must have ICDATA_WEBSERVICE_USER_ROLE role for the IC (context value). Please note global role cannot be used to access the webservice.

1.1.4. 1.4. Prerequisites

Institutes/Centers (ICs) need certificate-based authentication to access the ICDATA RESTful web service. For instructions on configuring the certificate, please refer to the System-to-System (S2S) Guide for eRA Web Services. As indicated in Section 3 of the Internal S2S Guide for eRA Web Services, the certificate needs to be registered after it has been procured. This step is conducted by the IC Coordinator. https://grants.nih.gov/grants/electronicreceipt/files/S2S_Web_Services_Guide.pdf

1.1.5. 1.5. Service Locations

Production: <https://services.internal.era.nih.gov/icdata/dataservices>

Stage: <https://services.internal.stage.era.nih.gov/icdata/dataservices>

Test: <https://services.internal.test.era.nih.gov/icdata/dataservices>

1.2. 2. ICDATA RESTful Web Services

The following Web Services should be used by IC Extension systems to update eRA application data.

The input payload data are expected to be in Query format and the output data will be in JSON format. Please ensure that **Content-Type** and **Accept** request headers are set correctly.

1.2.1. 2.1. Update Federal Person Involvement

This web service supports updating the fed_person_involvements_t table. For details on the structure of this table for query purposes, please refer to TechView documentation. Fed_person_involvements_t contains all external (i.e. PO, GMO, GS) staff involvements for a grant.

Converted from ice_dbms_fedp_a_pkg.assign_proc stored procedure.

1.2.1.1. 2.1.1. Webservice Parameters

applId

Mandatory. Identifies application record that change should be applied to. This is an inline parameter.

transactionType

Mandatory. Indicates type of transaction requested. Valid values are 'I' for insert, 'U' for update and 'D' for delete. Since there is only one record allowed for a given IC person involvement, in this procedure the

'I' and 'U' transaction types are treated the same. Either will create a new record if one does not exist, and overwrite an existing record if one does exist.

phsOrgCode

Mandatory. Identifies IC for which changes apply

roleTypeCode

Mandatory. Identifies type of involvement.

personId

Mandatory. Identifies the person to be assigned as the indicated role.

versionCode

Optional parameter. Default value for this parameter is 'M'

1.2.1.2. 2.1.2. Business Rules

1) As indicated in the procedure parameters, there can be only one PO per IC per grant. If an insert transaction is submitted when a record already exists, the existing record will be overwritten. Similarly, if an update transaction is submitted for a record that does not already exist, the record will just be created.

2) The personId specified must point to a person who is a valid Program Official for the indicated IC. A list of valid person_ids and their associated names for a given IC can be obtained using the following query:

```
SELECT person_id, last_name, first_name
FROM ic_potential_assignees
WHERE database_role_name = 'PROGRAM_OFFICIAL'
AND phs_org_code = < IC >
```

3) If the indicated phsOrgCode value does not exist in the icd_asgn_involvements_t table, it will automatically be added using the ice_dbms_icpcc_a_pkg.assign_ic_pcc_proc. All relevant rules for this procedure will apply.

4) When a PO record is removed, under no circumstances will a related record in icd_asgn_involvements_t be deleted, even if it was created automatically due to the business rule above. In this case the related record must be manually deleted by calling webservice endpoint /ice/ic-pcc/assign/{applId}.

5) All PO assignments made by the API will be considered 'manual' assignments. The API will create manual assignments in exactly the same way and using the same rules as the interactive Internal Referral Module (IRM) application. This means that assignments made by the API can overwrite assignments made by IRM and vice-versa. Manual assignments made by the API or IRM will take precedence over automated assignments created by assignment rules. In addition automated assignments made by the batch job will not overwrite manual assignments.

Web Attributes	Details
Endpoint	/ice/fed-person-involvement/assign/{applId}
Method	POST
Path Params	applId: Application ID
Request Query Parameters	transactionType : string /* Transaction type */ phsOrgCode : string /* Organization code */ roleTypeCode : string /* Role type code */ personId : integer /* Person ID */ versionCode : string /* Version code - Default value = 'M' */ Example: <pre>POST https://services.internal.era.nih.gov/icdata/dataservices/ice/fed-person-involvement/assign/12345?transactionType=U&phsOrgCode=GM&roleTypeCode=80&personId=8019957&versionCode=A</pre>
Success Response	Http Status: 200 Ok
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>

1.2.2. 2.2. Assign IC Involvement

Assign IC involvement for an application. Converted from ice_dbms_icpcc_a_pkg.assign_ic_pcc_proc stored procedure.

IC assignments are stored in icd_asgn_involvements_t and PCC assignments are stored in icd_prog_classes_t. For details on the structure of these two tables for query purposes, please refer to the TechView documentation. One change which should be noted from the current published TechView is that a new column, ic_main_flag, has been added to icd_prog_classes_t to indicate which record is the 'main' PCC. The valid values for this column are 'Y' and 'N'.

The ic-pcc/assign is a general-purpose procedure for adding, updating, or deleting IC and PCC assignments.

1.2.2.1. 2.2.1. Webservice Parameters

applId

Mandatory. Identifies application record that assignment change should be applied to. This is an inline parameter.

transactionType_

Mandatory. Indicates type of transaction requested. Valid values are 'I' for insert, 'U' for update and 'D' for delete.

phsOrgCode

Mandatory. Identifies IC to be added or deleted from application record.

oldPccCode

Optional - default NULL. For deletes, identifies old program class code to be deleted from application record. If a delete is specified with a NULL oldPccCode, the phsOrgCode assignment itself will be removed from the application. For updates, oldPccCode is used to specify the value to be replaced by newPccCode. For insert transaction the oldPccCode parameter is ignored.

newPccCode

Optional - default NULL. For inserts, identifies new program class code to be added to application record. If an insert is specified with a NULL newPccCode, the phsOrgCode will be assigned to the grant without any pcc assignment. For updates, newPccCode is used to specify the value to be used to replace oldPccCode. newPccCode cannot be NULL for update transactions. For delete transactions the newPccCode parameter is ignored.

Usage notes:

This procedure can be called to make IC and/or PCC assignment to grants. The following table illustrates the valid combinations of parameters for the procedure and the results of each. The table assumes that a phs_org_code of 'CA' has been passed to the procedure in all cases.

transactionType	old Pcc Code	new Pcc Code	Result
I	NULL	NULL	Adds CA to icd_asgn_involvements
D	NULL	NULL	Deletes all CA records from icd_asgn_involvements and icd_prog_classes_t.
I	NULL	XYZ	Assigns the XYZ PCC for CA in icd_prog_classes. If CA is not already in icd_asgn_involvements it will be automatically added.
U	ABC	XYZ	Replaces all occurrences of ABC with XYZ in icd_prog_classes and awd_fundings. If CA is not already in icd_asgn_involvements it will be automatically added.
D	ABC	NULL	Deletes all occurrences of ABC in icd_prog_classes and awd_fundings. However even if ABC is the only PCC assigned by CA, CA will not be automatically removed from icd_asgn_involvements. This must be done by a separate call to the procedure indicating a NULL new_pcc_code as specified above.

Web Attributes	Details
Endpoint	/ice/ic-pcc/assign/{applId}
Method	POST
Path Params	applId: Application ID
Request Query Parameters	<pre>transactionType : string /* Transaction type */ phsOrgCode : string /* Phs organization code */ oldPccCode : string /* Old program class code - Optional parameter - Default value is NULL */ newPccCode : string /* New program class code - Optional parameter - Default value is NULL */</pre> Example: <pre>POST https://services.internal.era.nih.gov/icdata/dataservices/ice/ic-pcc/assign/12345? transactionType=U&phsOrgCode=SM&oldPccCode=CN</pre>
Success Response	Http Status: 200 Ok

Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>
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1.2.3. 2.3. Assign Main PCC Involvement

Assign IC involvement for an application. Converted from ice_dbms_icpcc_a_pkg.assign_main_pcc_proc stored procedure.

1.2.3.1. 2.3.1. Webservice Parameters

applId

Mandatory. Identifies application record that assignment change should be applied to. This is an inline parameter.

phsOrgCode

Mandatory. Identifies IC to be added or deleted from application record

programClassCode

Mandatory. Identifies program class code to be assigned as the 'main' PCC for this phs_org_code_p.

overWriteFlag

Mandatory. If the overwrite parameter is set to 'Y', any existing IC main PCC for the indicated application will be overwritten with the specified new PCC in both icd_asgn_involvements_t and awd_fundings_t. If the overwrite parameter is set to 'N', any existing IC main PCC will be kept as a 'secondary' PCC.

1.2.3.2. 2.3.2. Business rules

1) IC assignments can only be made for competing applications: appl_type_code = 1, 2, 9 or (3 or 4 with council_meeting_date not '%00' and council_meeting_date not null).

2) When an IC assignment is made, it will become the 'dual' assignment if no dual exists and the grant has not been awarded (appl_status_code not in (3,5,6,19,20). Dual assignments are currently indicated by icd_asgn_involvements_t.legacy_source_master_item = '004'

. 3) Cannot remove the administering IC.

4) Cannot remove the IC marked as 'dual' if dual actions have been entered (appls_t.dual_recom_code is not null).

5) Cannot remove any IC that has provided funding.

6) If there is one or more PCCs assigned by an IC, one will always be marked as 'main'. The API does this by default whether or not the assign_main_pcc_proc is used. For example, if no PCC exists and the assign_ic_pcc_proc is used to create one, it will be marked as 'main'. Similarly, if two PCCs exist and the 'main' one is deleted by assign_ic_pcc_proc, the remaining PCC will automatically be marked as 'main'. The 'main' program class code has special significance in IMPAC II:

a) The main program class code is the one assigned by the program class code auto-assignment batch module. If a manually assigned main program class code exists, the auto-assignment batch module will not make any assignments for this grant/IC.

b) The main program class code is the one used by the GMO/GS/PO auto-assignment batch module when applying assignment rules to grants.

c) For the administering org, the program class code indicated as main will also be recorded in appls_t.prog_class_code.

7) All PCC assignments made by the API will be considered 'manual' assignments. The API will create manual assignments in exactly the same way and using the same rules as the interactive Internal Referral Module (IRM) application. This means that assignments made by the API can overwrite assignments made by IRM and vice-versa. Manual assignments made by the API or IRM will take precedence over automated assignments created by assignment rules. For example, if an assignment rule exists which makes the assignment batch job associate PCC 'X' as the IC main PCC for an application, but the API is then called to assign PCC 'Y' as main, the 'Y' will overwrite the 'X'. In addition subsequent runs of the batch job will not overwrite the manually assigned 'Y'. It should be noted that the batch module only deals with the IC main PCC assignment for each application.

8) All PCCs assigned must be valid based on the range of valid values defined for the relevant IC in the prog_classes_t table. The procedure will use the exact same validation logic as is currently used by the Grants Management edit checker.

Web Attributes	Details
Endpoint	/ice/main-pcc/assign/{applId}
Method	POST
Path Params	applId: Application ID
Request Query Parameters	phsOrgCode : String /* Phs organisation code */ progClassCode : String /* This is for setting status to Activated */ overWriteFlag : string /* Over write flag. Default value "Y" */ Example: <pre>POST https://services.internal.era.nih.gov/icdata/dataservices/ice/main-pcc/assign/12345? phsOrgCode=SM&progClassCode=SM&overWriteFlag=Y</pre>
Success Response	Http Status: 200 OK
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>

1.2.4. 2.4. Assign Award Funding

Update tables in accordance with various business rules. Converted from ice_dbms_awfd_a_pkg.assign_proc stored procedure.

1.2.4.1. 2.4.1. Webservice parameters

applId

Mandatory. Identifies application record that change should be applied to. This is an inline parameter.

transType

Mandatory. Indicates type of transaction requested. Valid values are 'I' for insert, 'U' for update and 'D' for delete.

phsOrgCode

Mandatory. Identifies IC for which changes apply

oldCan

Optional - default NULL. For deletes, identifies old CAN to be deleted from application record. For insert transactions the oldCan parameter must be omitted or set to NULL.

newCan

Optional – default 'SAME'. For inserts, identifies new CAN to be inserted. For updates, identifies value oldCan should be replaced with. If procedure is being called to update PCC but not CAN, this parameter should be omitted, set to the keyword 'SAME', or set equal to the value specified in oldCan. For deletes this parameter must be omitted or set to NULL as it is not relevant.

newPcc

Optional - default 'SAME'. For inserts, identifies new program class code to be associated with newCan For updates, newPcc is used to specify the value to be associated with the current record that contains oldCan. newPcc can be set to NULL to blank out the program class code associated with oldCan. If procedure is being called to update CAN but not PCC, this parameter should be omitted or set to the keyword 'SAME'. For deletes this parameter must be omitted or set to NULL as it is not relevant.

Usage notes:

The following table illustrates the valid combinations of parameters for the procedure and the results of each. The table assumes that a valid appl_id and phs_org_code have been passed to the procedure in all cases.

transType	oldCan	newCan	newPcc	Result
-----------	--------	--------	--------	--------

I	NULL	1234567	NULL	Adds a record with CAN 1234567 and NULL prog_class_code to awd_fundings_t
I	NULL	1234567	XYZ	Adds a record with CAN 1234567 and prog_class_code XYZ to awd_fundings_t
U	1234567	7654321	SAME	Updates the record with CAN 1234567 to CAN 7654321, and does not change the prog_class_code.
U	1234567	SAME	XYZ	Updates the record with CAN 1234567 to have a prog_class_code of XYZ, and does not change the CAN
U	1234567	7654321	XYZ	Updates the record with CAN 1234567 to CAN 7654321, and changes the prog_class_code to XYZ
U	1234567	SAME	NULL	Updates the record with CAN 1234567 to NULL out the prog_class_code, and does not change the CAN.
D	1234567	NULL	NULL	Deletes the record with CAN 1234567.

1.2.4.2. 2.4.2. Business Rules

- 1) The CAN specified must be a valid CAN in the cans_t table for the relevant IC.
- 2) If an award has been prepared (appl_status_code in (3,5,6,19,20)), the CAN specified must be an active CAN (active_code = 'Y') in the cans_t table for the relevant IC.
- 3) If an award has been prepared, only update transactions can be performed (i.e. insert and delete transactions are not allowed).
- 4) If the grant is considered work-in-progress by the Grants Management Award system, the CAN cannot be updated, inserted, or deleted.
- 5) If an award had been prepared, cofunded CANs cannot be updated. This is a temporary restriction until reverse bridge issues concerning cofunded CANs can be worked out.
- 6) If an award has been prepared, the CAN can be updated only for grants whose fiscal year is the current fiscal year.
- 7) If an award has been prepared, only the last 4 digits of the CAN can be updated.
- 8) If an award has been prepared, any valid CAN change will be propagated to OFM the next day. CAN assignments made prior to an award being prepared will not be propagated to OFM until the grant is awarded.
- 9) When inserting a CAN record, only 'BUD' awd_funding records will be created. In order for a CAN record to be inserted, the 'BUD' period record must already exist in the appl_periods_t table. The ice_dbms_appd_a_pkg can be used to create record in appl_periods_t.
- 10) When updating a CAN record, all awd_funding records (i.e. 'BUD', 'COM', etc.) for that CAN will be updated except for those that are 'work in progress' (i.e. 'WBUD', 'WCOM', etc.).
- 11) When deleting a CAN record, all awd_funding records (i.e. 'BUD', 'COM', etc.) for that CAN will be deleted except for those that are 'work in progress' (i.e. 'WBUD', 'WCOM', etc.).
- 12) CAN changes made to awd_fundings will automatically be propagated to the appl_budget_items_t child table.
- 13) If the indicated phs_org_code_p value does not exist in the icd_asgn_involvements_t table, it will automatically be added using the ice_dbms_icpcc_a_pkg.assign_ic_pcc_proc. All relevant rules for this procedure will apply.
- 14) If the indicated new_pcc_p value does not exist in the icd_prog_classes_t table, it will automatically be added using the ice_dbms_icpcc_a_pkg.assign_ic_pcc_proc. All relevant rules for this procedure will apply.
- 15) When a CAN record is removed, under no circumstances will any related records in icd_asgn_involvements_t or icd_prog_classes_t be deleted, even if they were created automatically due to one of the business rules above. In this case these records must be manually deleted by calling the webservice endpoint /ice/ic-pcc/assign/{applId}..
- 16) If 'child' records exist for a CAN to be deleted, the delete will be rejected due to a database constraint. Currently the only possible child records are in the appl_budget_items_t table. In order to delete a CAN that has associated budget item details, the details must be deleted first. Currently budget item details can only be deleted through the Grants Management application.

Web Attributes	Details
Endpoint	/ice/awd-fundings/assign/{applId}
Method	POST
Path Params	applId: Application ID

Request Query Parameters	<pre> transType : string /* Transaction type */ phsOrgCode : string /* Phs organization code */ oldCan : string /* Old CAN - This is an optional parameter */ newCan : string /* New CAN - This is an optional parameter*/ newPcc : string /* New program class code - This is an optional parameter */ Example: POST https://services.internal.era.nih.gov/icdata/dataservices/ice/awd-fundings/assign/12345?transType=U&phsOrgCode=SM&oldCan=8472428&newPcc=HEJ </pre>
Success Response	Http Status: 200 OK
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre> { "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] } </pre>

1.2.5. 2.5. Assign Appl Period Funding

Update tables in accordance with various business rules. Converted from ice_dbms_appd_a_pkg.assign_proc stored procedure.

1.2.5.1. 2.5.1. Webservice parameters

applID

Mandatory. Identifies application record that change should be applied to.

transType

Mandatory. Indicates type of transaction requested. Valid values are 'I' for insert or 'U' for update. Since there is only one budget period record allowed for a given grant, in this procedure the 'I' and 'U' transaction types are treated the same. Either will create a new record if one does not exist, and overwrite an existing record if one does exist.

phsOrgCode

Mandatory. Identifies IC for which changes apply

totalPeriodAmt

Optional - default NULL. Total amount projected to fund the grant for the current period. This value may be EMPTY if the desire is to 'delete' previously assigned dollar figures.

directCostAmt

Optional - default NULL. Direct costs projected for the current period. This value may be EMPTY if the desire is to 'delete' previously assigned dollar figures or if the direct/indirect cost breakdown is not known.

indirectCostAmt

Optional - default NULL. Indirect costs projected for the current period. This value may be EMPTY if the desire is to 'delete' previously assigned dollar figures or if the direct/indirect cost breakdown is not known.

1.2.5.2. 2.5.2. Business Rules

1) For consistency with other procedures, the phs_org_code_p parameter is included. However, only transactions for records where the user has access to the administering IC (appls_t.admin_phs_org_code) are supported.

2) Delete transactions are not supported by this procedure, as the relevant appl_period records may also contain other information assigned outside the scope of this procedure (such as period dates). However, any dollar figures supplied using this procedure may be removed by performing an update transaction with all NULL values.

- 3) Only the 'BUD' period may be inserted or updated using this procedure. The appl_period_num associated with the BUD period is always the year of support (indicated in appls_t.support_year).
- 4) A total cost may be given for the BUD period, without any breakdown into direct and indirect costs (i.e. the direct_cost_amt_p and indirect_cost_amt_p parameters may be NULL). However, if direct/indirect costs are given for the BUD period, then the total cost must equal the sum of the direct and indirect costs.
- 5) As is the case with the intent_to_pay_code, the BUD period cannot be inserted or updated when the appl_status_code = '17' (pending review). Also, the BUD period cannot be inserted or updated once the grant award has been prepared (appl_status_code in (3,5,6,19,20)).
- 6) If the grant is considered work-in-progress by the Grants Management Award system, no transactions can be processed by this procedure.
- 7) As indicated in the procedure parameters, there can be only one budget period per grant. If an insert transaction is submitted when a record already exists, the existing record will be overwritten. Similarly, if an update transaction is submitted for a record that does not already exist, the record will just be created.

Web Attributes	Details
Endpoint	/ice/appd-fundings/assign/{applId}
Method	POST
Path Params	applId: Application ID
Request Query Parameters	transType : string /* Transaction type */ phsOrgCode : string /* Phs organization code */ totalPeriodAmt : Integer/* Total Period Amount - This is an optional parameter */ directcostAmt : Integer/* Direct Cost Amount - This is an optional parameter*/ indirectcostAmt: Integer /* Indirect Cost Amount - This is an optional parameter */ Example: <pre>POST https://services.internal.era.nih.gov/icdata/dataservices/ice/appd-fundings/assign/12345? transType=U&phsOrgCode=CA&totalPeriodAmt=246916&directCostAmt=123457&indirectCostAmt=123457 -- for empty/null parameters - totalPeriodAmt, directcostAmt, indirectcostAmt POST https://services.internal.era.nih.gov/icdata/dataservices/ice/appd-fundings/assign/12345? transType=U&phsOrgCode=CA&totalPeriodAmt=&directCostAmt=&indirectCostAmt=</pre>
Success Response	Http Status: 200 OK
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>

1.2.6. 2.6. Get Program Class Code Related Details

API to get program class code and related details for the given appl Id and Phs Org Code.

1.2.6.1. 2.6.1. Webservice parameters

applID

Mandatory. Identifies application record that change should be applied to.

phsOrgCode

Mandatory. Identifies IC for which changes apply

1.2.6.2. 2.6.2. Business Rules

NA

Web Attributes	Details
Endpoint	/ice/pcc-details/{applId}
Method	GET
Path Params	applId: Application ID
Request Query Parameters	phsOrgCode : string /* Phs organization code */ Example: GET https://services.internal.era.nih.gov/icdata/dataservices/ice/pcc-details/12345?phsOrgCode=CA
Success Response	Http Status: 200 OK { "applId": 9483447, "phsOrgCode": "CA", "applIdProgClassDetails": [{"progClassCode": "X3BP", "icMainFlag": "Y", "asgnVersionCode": "M"}] }
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>

1.2.7. 2.7. Get Notification details

API to retrieve notification data for given notification types. The mapped user must have **ICDDATA_NOTIFICATION_WEBSERVICE_USER_ROLE**

1.2.7.1. 2.7.1. Webservice parameters

notificationTypes

Mandatory. The notification type(s) comma separated values

lastRunDt

Optional. Default CurrentDate. The date/time after which notification data is required

1.2.7.2. 2.7.2. Business Rules

Allowed Notification Types : FFR_REJECTION, FRAM_SUBMITTED_INTERNAL, GCM_CLOSEOUT_COMPLETE_LETTER, FPR_SUBMITTED_INTERNAL, FPR_SUBMITTED_EXTERNAL, GCM_GCC_LTR1, GCM_GCC_LTR2, GCM_GCC_LTR3

Web Attributes	Details
Endpoint	/ice/notification-data
Method	GET

Request Query Parameters	notificationTypes : string /* The notification type(s) comma separated values. This is mandatory */ lastRunDt : string /* The lastRunDt in MM/dd/yyyy hh:mm:ss format */ Example: <pre>GET https://services.internal.stage.era.nih.gov/icdata/dataservices/ice/notification-data?notificationTypes=FFR_REJECTION&lastRunDt=01/11/2020 10:10:09</pre>
Success Response	Http Status: 200 OK <pre>[{"eventLogId":38669678,"eventKeyId":9538040,"notificationName":"FFR_REJECTION","createdDate":"01/27/2020T11:27:35-0500"}, {"eventLogId":38669667,"eventKeyId":9612513,"notificationName":"FFR_REJECTION","createdDate":"01/27/2020T11:10:45-0500"}, {"eventLogId":38664146,"eventKeyId":9430394,"notificationName":"FFR_REJECTION","createdDate":"01/21/2020T11:03:55-0500"}]</pre>
Error Response	Http Status: 400 for business exceptions, 500 for unknown server errors Error Payload type: application/json <pre>{ "businessErrorCode" : [Error Code], "businessErrorMessage" : [Error Message] }</pre>

1.3. 3. Exceptions and Codes

Following is a list of exceptions which may be raised by the API procedures. Except as noted most exceptions are probably the result of calling a procedure incorrectly or violating the business rules. If it appears the procedure is being called correctly and an exception is still encountered, send the error number and message to the IMPAC II helpdesk along with the parameters passed to the procedure.

-20801 to -20809 Security based exceptions

-20801 No module name recorded

The application making the procedure call has not recorded its module name using thedbms_application_info.set_module procedure as required.

-20802 Recorded module name [X] not registered

The module name recorded by the application making the procedure call has not been registered with IMPAC II. Contact the IC coordinator to have the application registered.

-20803 Registered module name [X] does not begin with ICE

The registered module name does not begin with the required ICE acronym.

-20804 User [X] does not have access to IC [Y];

The X user does not have privileges to make assignments for IC Y. Contact the IC coordinator for the IC in question and have them grant privilege to user X.

-20810 to -20819 Global exceptions

-20810 Invalid combination of NULL/NOT NULL parameters specified

A mandatory parameter was left NULL, or a NOT NULL value was specified in a parameter which should be NULL. Refer to Section 2 for valid procedure parameter settings.

-20811 Appl_id [X] not valid

The appl_id specified does not exist.

-20812 Trans_type [X] not valid

The specified trans_type is not valid for this procedure. The valid trans_types for most procedures are 'I' (insert), 'U' (update), and 'D' (delete). As specified in the business rules, certain procedures do not support all three of these transaction types.

-20813 Internal procedure exception

This code represents an internal exception in the procedure. The corresponding message will further describe the error. If this error is encountered it should be sent to the IMPAC II helpdesk along with the parameters used.

-20814 Other Oracle error

This code will be used for any other Oracle error which is encountered that is not specifically caught by the procedure. The corresponding message will further describe the error. In some cases this error may also be a result of user error, or there may be an overall system problem. If this error is encountered it should be sent to the IMPAC II helpdesk along with the parameters used.

-20820 to -20829 Assign IC Involvement (Endpoint: /ice/ic-pcc/assign/{applId}) specific exceptions

-20820 PCC [X] not found

An update was requested of PCC X, but this PCC is not currently assigned to the grant indicated.

-20822 PCC [X] not valid for IC [Y]

The PCC indicated is not valid based on the business rules defined for the indicated IC. The business rules for PCC validations use predefined logic defined for each IC based on the prog_classes_t table. Contact the IC coordinator if the PCC indicated should be made valid.

-20823 Cannot delete primary org involvement

An attempt was made to delete the administering org involvement on the grant.

-20824 Cannot delete dual org when dual actions have been entered

An attempt was made to delete the dual org involvement on the grant when dual actions have been entered.

-20825 Cannot delete org with funding

An attempt was made to delete an org involvement that provided funding for the grant.

-20826 Cannot make IC assignment for noncompeting grant

An attempt was made to add an IC involvement to a noncompeting grant.

20830 to -20840 Assign Award Funding (Endpoint: /ice/awd-fundings/assign/{applId}) specific exceptions

-20830 CAN [X] is not valid for IC [Y]

The CAN specified is not present in the cans_t lookup table for the specified IC.

-20831 CAN [X] is not active

The CAN specified is present in the cans_t lookup table for the specified IC, but is marked as inactive. Only active CANs can be assigned to grants where the award has been prepared.

-20832 Cannot insert or delete records when an award has been prepared

An attempt was made to perform an insert or delete transaction on a grant where the award has been prepared.

-20833 Can only update CANs for current fiscal year

An attempt was made to update the CAN on a grant where the award had been prepared for a previous fiscal year.

-20834 Can only update last 4 digits of awarded CAN

An attempt was made to update one or more of the first 3 digits on a grant where the award has been prepared.

-20835 CAN [X] not found on grant N

An attempt was made to update or delete a CAN that is not assigned to the specified grant.

-20836 CAN not updateable because GM work-in-progress exists for this grant. To change CAN, first delete WIP in Grants Mgmt. Award System.

An attempt was made to update, insert, or delete a CAN on a grant which is currently being processed by the Grants Management Award system.

-20839 CAN not updateable on cofunded grants.

An attempt was made to update a CAN on a cofunded grant. At this time this type of update is not supported.

-20840 Cannot insert funding - no budget record exists in appl_periods_t.

An attempt was made to insert a record when there is no supporting 'BUD' record in appl_periods_t. ice_dbms_appd_a_pkg can be used to created 'BUD' records in appl_periods_t.

-20841 to -20849 Update Federal Person Involvement (Endpoint: /ice/fed-person-involvement/assign/{applId}) specific exceptions

-20841 Person specified [N] is not a valid X

The person_id specified is not a valid candidate for the specified role and IC. For a list of valid candidate persons for a role, query the ic_potential_assignees view as indicated under the business rules.

-20842 Role [X] is not supported by API

The role specified cannot be modified by the API.

-20843 Specified involvement does not exist

An attempt was made to delete an involvement record that does not exist.

-20850 to -20859 other exceptions

-20850 IC indicated [X] is not administering org of grant

The specified IC is not the administering org of the grant. Only the administering org can update the appl record.

-20852 Cannot update grant pending review

An attempt was made to update a grant pending review.

-20853 Indicated intent to pay value [X] is not valid

The specified intent to pay code valid is not valid.

-20860 to -20869 other specific exceptions

-20860 Only administering org can update item [X]

The specified IC is not the administering org of the grant. Only the administering org can update the item indicated.

-20861 Only dual org can update item [X]

The specified IC is not the dual org of the grant. Only the dual org can update the item indicated.

-20862 Indicated item code [X] is not valid

The indicated item_code is not a valid gen_data entry. Only items 801,802,803,804 and 805 are valid.

-20863 Specified item code [X] does not exist

An attempt was made to delete an gan_data item record that does not exist.

-20870 to -20879 Assign Appl Period Funding (Endpoint: /lice/appd-fundings/assign/{applId}) specific exceptions

-20870 IC indicated [X] is not administering org of grant

The specified IC is not the administering org of the grant. Only the administering org can update the appl record.

-20871 Cannot insert or delete records when an award has been prepared

An attempt was made to perform an insert or delete transaction on a grant where the award has been prepared.

-20872 Budget not updateable because GM work-in-progress exists for this grant. To change budget, first delete WIP in Grants Mgmt. Award System.

An attempt was made to update, insert, or delete a CAN on a grant which is currently being processed by the Grants Management Award system.

-20873 Cannot update grant pending review

An attempt was made to update a grant pending review.

-20874 Total cost does not equal sum of direct and indirect

A cost breakdown is indicated for direct and indirect costs but the total period amount indicated is not the sum of these figures.

1.4. 4. Web Services Definition

icdata.yaml

```
swagger: "2.0"
info:
  version: 1.0.0
  title: IC Data API
```

description: A set of APIs used by IC Extension systems to update application specific information.
host: services.internal.era.nih.gov
basePath: /icdata/api
schemes:
- https

paths:

/ice/fed-person-involvement/assign/{applId}:
post:
description: Update Federal Person Involvement records for an application. Converted from ice_dbms_fedp_a_pkg.assign_proc stored procedure.
operationId: assignFedPersonInvolvement
produces:
- application/json
parameters:
- name: applId
in: path
description: ID of the target application.
required: true
type: integer
- name: transactionType
in: query
description: Transaction type
required: true
type: string
- name: phsOrgCode
in: query
description: Phs organization code
required: true
type: string
- name: roleTypeCode
in: query
description: Role type code
required: true
type: string
- name: personId
in: query
description: Person ID
required: true
type: integer
- name: versionCode
in: query
description: Version code. Default value = 'M'.
required: false
type: string
responses:
200:
description: Successful response
400:
\$ref: "#/responses/BadRequest"
401:
\$ref: "#/responses/UnauthorizedRequest"
default:
\$ref: "#/responses/BadRequest"

/ice/ic-pcc/assign/{applId}:
post:
description: Assign IC involvement for an application. Converted from ice_dbms_icpcc_a_pkg.assign_ic_pcc_proc stored procedure.
operationId: assignICInvolvement
produces:
- application/json
parameters:
- name: applId
in: path
description: ID of the target application.
required: true
type: integer
- name: transactionType
in: query

```

        description: Transaction type
        required: true
        type: string
    - name: phsOrgCode
      in: query
      description: Phs organization code
      required: true
      type: string
    - name: oldPccCode
      in: query
      description: Old Program Class Code
      required: false
      type: string
    - name: newPccCode
      in: query
      description: New Program Class Code
      required: false
      type: string
  responses:
    200:
      description: Successful response
    400:
      $ref: "#/responses/BadRequest"
    401:
      $ref: "#/responses/UnauthorizedRequest"
    default:
      $ref: "#/responses/BadRequest"

/ice/main-pcc/assign/{applId}:
  post:
    description: Assign IC involvement for an application. Converted from ice_dbms_icpcc_a_pkg.
    assign_main_pcc_proc stored procedure.
    operationId: assignMainPccInvolvement
    produces:
      - application/json
    parameters:
      - name: applId
        in: path
        description: ID of the target application.
        required: true
        type: integer
      - name: phsOrgCode
        in: query
        description: Phs organization code
        required: true
        type: string
      - name: progClassCode
        in: query
        description: Program Class Code
        required: true
        type: string
      - name: overWriteFlag
        in: query
        description: Overwrite Flag. Default value "Y"
        required: false
        type: string

  responses:
    200:
      description: Successful response
    400:
      $ref: "#/responses/BadRequest"
    401:
      $ref: "#/responses/UnauthorizedRequest"
    default:
      $ref: "#/responses/BadRequest"

/ice/awd-fundings/assign/{applId}:
  post:
    description: Update tables in accordance with various business rules. Converted from
    ice_dbms_awfd_a_pkg.assign_proc stored procedure.

```

```

operationId: assignAwardFunding
produces:
  - application/json
parameters:
  - name: applId
    in: path
    description: ID of the target application.
    required: true
    type: integer
  - name: transType
    in: query
    description: Transaction Type
    required: true
    type: string
  - name: phsOrgCode
    in: query
    description: Phs organization code
    required: true
    type: string
  - name: oldCan
    in: query
    description: Old CAN
    required: false
    type: string
  - name: newCan
    in: query
    description: New CAN
    required: false
    type: string
  - name: newPcc
    in: query
    description: New Program Class Code
    required: false
    type: string
responses:
  200:
    description: Successful response
  400:
    $ref: "#/responses/BadRequest"
  401:
    $ref: "#/responses/UnauthorizedRequest"
  default:
    $ref: "#/responses/BadRequest"
/ice/appd-fundings/assign/{applId}:
  post:
    description: Assign IC appl period funding for an application. Converted from ice_dbms_appd_a_pkg.
    assign_proc stored procedure.
    operationId: assignApplPeriodFunding
    produces:
      - application/json
    parameters:
      - name: applId
        in: path
        description: ID of the target application.
        required: true
        type: integer
      - name: transType
        in: query
        description: Transaction Type
        required: true
        type: string
      - name: phsOrgCode
        in: query
        description: Phs organization code
        required: true
        type: string
      - name: totalPeriodAmt
        in: query
        description: Total Period Amount
        required: false
        type: integer

```



```

    - name: directCostAmt
      in: query
      description: Direct Cost Amount
      required: false
      type: integer
    - name: indirectCostAmt
      in: query
      description: Indirect Cost Amount
      required: false
      type: integer

  responses:
    200:
      description: Successful response
    400:
      $ref: "#/responses/BadRequest"
    401:
      $ref: "#/responses/UnauthorizedRequest"
    default:
      $ref: "#/responses/BadRequest"
/ice/pcc-details/{applId}:
  get:
    description: Fetch IC-PCC details for an application.
    operationId: getPcc
    produces:
      - application/json
    parameters:
      - name: applId
        in: path
        description: ID of the target application.
        required: true
        type: integer
      - name: phsOrgCode
        in: query
        description: Phs organization code
        required: true
        type: string
    responses:
      '200':
        description: OK
        schema:
          $ref: "#/definitions/ResponsesIcdProgClass"
      400:
        $ref: "#/responses/BadRequest"
      401:
        $ref: "#/responses/UnauthorizedRequest"
      default:
        $ref: "#/responses/BadRequest"

responses:
  BadRequest:
    description: Invalid parameters or business error
    schema:
      $ref: "#/definitions/ApiError"
  UnauthorizedRequest:
    description: Unauthorized Request error
    schema:
      $ref: "#/definitions/ApiError"
  IcdProgClass:
    description: IC PCC Data
    schema:
      $ref: "#/definitions/ResponsesIcdProgClass"

definitions:
  ApiError:
    type: object
    description: Error details
    properties:
      businessErrorCode:
        type: integer
      businessErrorMessage:

```

```
      type: string
    ApplIcdProgClassDetail:
      type: object
      properties:
        progClassCode:
          type: string
        icMainFlag:
          type: string
        asgnVersionCode:
          type: string
    ResponsesIcdProgClass:
      type: object
      properties:
        applId:
          type: integer
        phsOrgCode:
          type: string
        applIcdProgClassDetails:
          type: array
          items:
            $ref: "#/definitions/ApplIcdProgClassDetail"
```